Short circuits and fuses — Part 1 12a

In an electric circuit some electrical energy needs to be used up; that is, a resistance must slow the electric current, otherwise the unusually large amount of current travelling through the circuit will cause some of the circuit components to heat up, to melt or to spark. When components melt or get wet, electricity will flow along the path of least resistance, possibly causing a short circuit which often leads to fire. Fuses are designed to 'blow' before any item in a circuit causes a fire. When a fuse blows, the circuit is interrupted and electricity no longer flows.

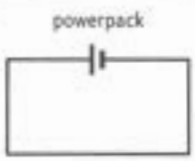
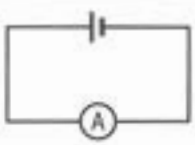
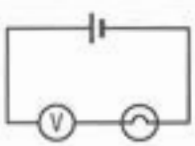
Ammeters measure the current. Since they have a very low resistance themselves and do not influence the current, they are switched in series with the item to be measured. Voltmeters, on the other hand, measure the potential difference and do not want any current to flow through them. Therefore, a voltmeter has a very high resistance and is switched parallel with the item to be measured.

1 Summarise the information about ammeters and voltmeters by completing the table below.

Aspect	Voltmeters	Ammeters
Measures what?		
What does this measurement mean?		
Measures in what unit?		
Abbreviation for unit of measurement?		
Has what level of resistance?		
Is switched how?		
Is current allowed to travel through?		

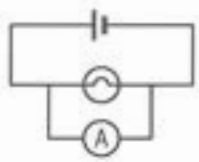
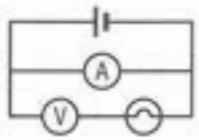
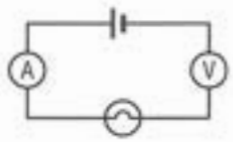
Fixing electrical circuits

2 The following circuits all have something wrong with them, causing a short circuit. Describe what has caused the short circuit in each circuit and suggest, if possible, how to rectify the problem.

Circuit	Description of problem	Suggestion for rectification
a 		
b 		
c 		

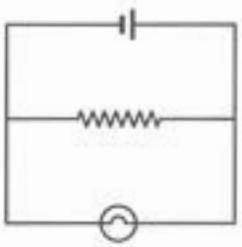
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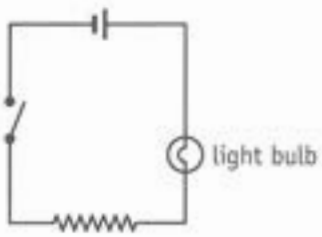
Short circuits and fuses — Part 1 12b

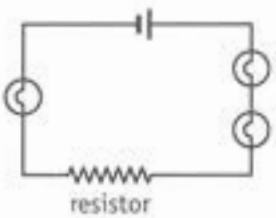
Circuit	Description of problem	Suggestion for rectification
d 		
e 		
f 		

Measuring electricity

3 Add ammeters and voltmeters to measure the current and potential difference of the indicated items. Describe each circuit in your own words.

a 

b 

c 

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